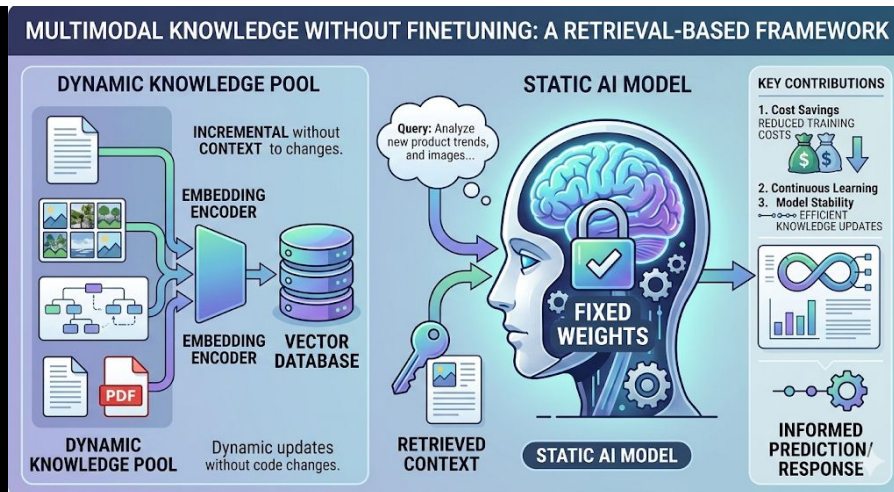


A Retrieval-Based Multi-Modal Learning Framework for Incremental Knowledge Integration Without Model Fine-Tuning



Presented By-

Sk. Roushan Khalid 221-35-877

Supervisor: Md. Shohel Arman

Assistant Professor, Department of Software Engineering, Daffodil Int. University

Outlines



Research Gaps and Problem Statement



Objectives and Contribution



Research Scope and Literature Review



Methodology



References



Publications

Research Gaps & Problem Statement

Research Gap: Current AI research mostly uses traditional model fine-tuning, which costs a lot of computing power because it needs constant retraining. This ignores very efficient retrieval methods that don't need any training.

Problem Statement: Many organizations can't afford to retrain models because they lack powerful hardware (like strong GPUs) and have to deal with huge amounts of data. This makes traditional retraining too difficult and expensive. Therefore, we need a flexible system based on retrieval that can quickly add new knowledge without having to change the main model, saving resources.

References:

1. [Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification](#)
2. [RETRIEVAL-ENHANCED CONTRASTIVE VISION-TEXT MODELS](#)

Objective

- **Creating a hardware-friendly AI framework:** Building a system that instantly learns from new images and text without requiring expensive GPUs or constant retraining.
- **Designing a self-cleaning memory:** Developing practical techniques to automatically filter out bad data and organize the vector database, keeping predictions fast and accurate.
- **Proving the cost-efficiency:** Demonstrating that a simple "search-and-match" classification approach is a realistic, budget-friendly alternative to traditional AI model building for real-world businesses.

Research Scope

- **Handling text and images:** The study will focus specifically on processing and matching multimodal data like images and text descriptions, leaving out video or audio for now.
- **Using only frozen models:** The research will strictly use existing, pre-trained AI encoders (like CLIP) without modifying their weights or doing any GPU-heavy fine-tuning.
- **Focusing on database cleanup:** The work will explore specific techniques to filter out bad data, handle conflicting labels, and organize the vector memory so the system searches faster and smarter.
- **Comparing cost vs. accuracy:** The evaluation will be limited to measuring how much hardware cost and processing time this "search-and-match" method saves compared to traditional model training on a specific test dataset.

Literature Review



Author	Year	Paper Name	Methods	Findings	Gaps
Zhang et al.	2022	Tip-Adapter: Training-free adoption of CLIP...	Uses a Key-Value Cache model to store few-shot visual features as keys and labels as values.	Achieved competitive accuracy without any SGD training or backpropagation.	Performance is static; if the cache has "noisy" or bad data, the accuracy drops immediately.
Yasunaga et al.	2023	Retrieval-Augmented Multimodal Language Modeling	RA-CM3 framework: Retrieves multimodal documents to assist in generative tasks.	External knowledge significantly reduces "hallucinations" in vision-language tasks.	Focused on generation, not efficient classification or low-resource hardware optimization.
Banani et al.	2024	Retrieval-Enhanced Contrastive Vision-Text Models	RECO framework: Refines CLIP embeddings using cross-modal retrieved info at inference.	Massive gains (+10.9% on Stanford Cars) for fine-grained classification.	Requires high-quality external memory; lacks a mechanism to self-clean the database.
Gui et al.	2024	kNN-CLIP: Retrieval enables training-free segmentation...	k-Nearest Neighbor retrieval from a continually expanding database of instance embeddings.	Prevents catastrophic forgetting by storing knowledge non-parametrically.	Segmentation is computationally heavy; doesn't focus on the "budget-friendly" business case.
Wang et al.	2024	A Hard-to-Beat Baseline for Training-free CLIP...	Applies Gaussian Discriminant Analysis (GDA) to estimate class distributions without training.	Surpassed SOTA methods in imbalanced and out-of-distribution (OOD) data tasks.	Assumes data follows a Gaussian distribution, which fails with unorganized/messy real-world data.

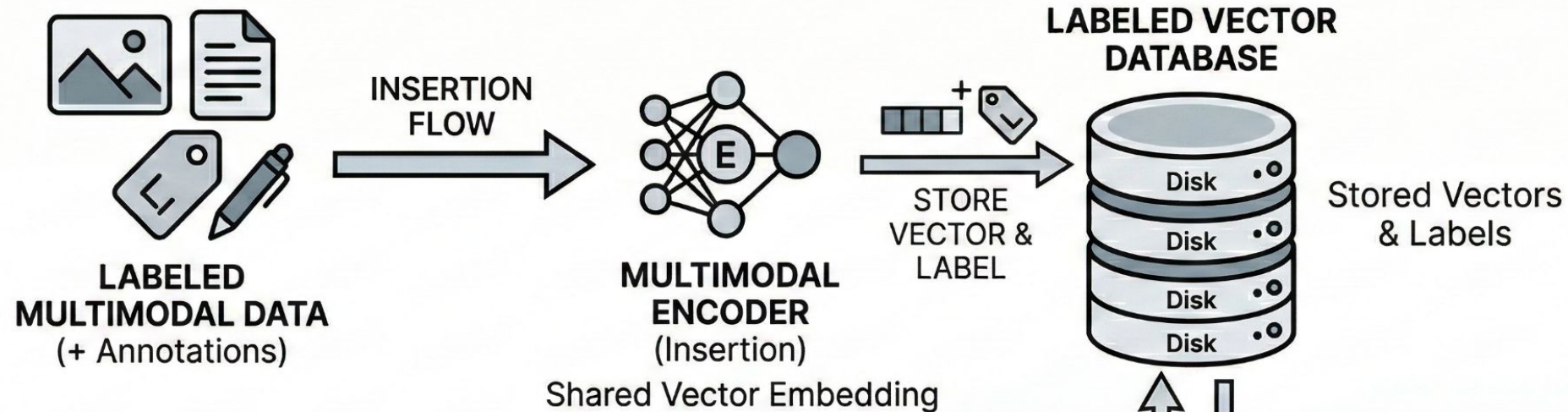
Literature Review



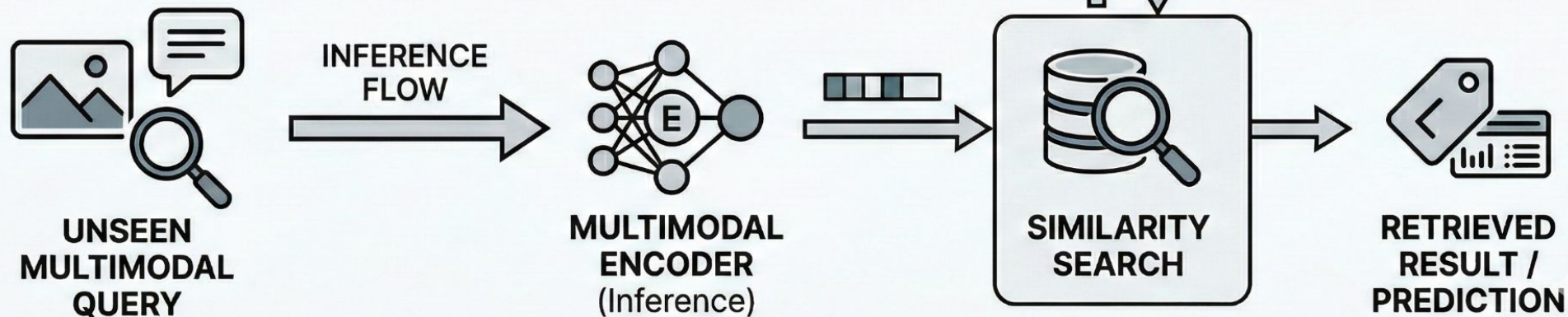
Author	Year	Paper Name	Methods	Findings	Gaps
Li et al.	2024	Text and Image Are Mutually Beneficial	TIMO framework: Dual-pathway guidance where text rectifies image matches and vice-versa.	Surpassed traditional training-required methods with 100x less time cost.	Still suffers from "anomalous matches" if the input query is significantly degraded or noisy.
Wang et al.	2024	Proto-Adapter: Efficient Training-Free CLIP-Adapter...	Stores Class Prototypes (mean vectors) instead of every individual sample in memory.	Drastically reduced memory size compared to Tip-Adapter while maintaining accuracy.	Using "means" loses fine-grained diversity within a class; needs smarter database organization.
Mitra et al.	2023	Effect of Pivot Language and Segment-Based Few-Shot...	Pivot-language RAG: Uses high-resource language embeddings to help low-resource tasks.	Efficient retrieval improved intent identification in low-resource NLU settings.	Limited to text-only cross-domain transfer; doesn't address the vision-text "modality gap."
Baumann et al.	2024	Combining RAG and Few-Shot Learning for Model Synthesis...	Pairs RAG with FSL to help LLMs understand uncommon Domain Specific Languages (DSLs).	RAG identified the best examples for the prompt, making the model "smarter" instantly.	Focused on software modeling; lacks the "self-cleaning" logic for vector databases.
Ashok & Lipton	2024	Improving Few-Shot Cross-Domain NER...	Instruction Tuning with word-level embeddings to retrieve in-prompt examples for NER.	Outperformed previous SOTA in cross-domain Named Entity Recognition tasks.	High dependency on GPT-4 level backbones, which contradicts your "budget-friendly" goal.

Methodology

DATABASE ENHANCEMENT (INSERTION FLOW)



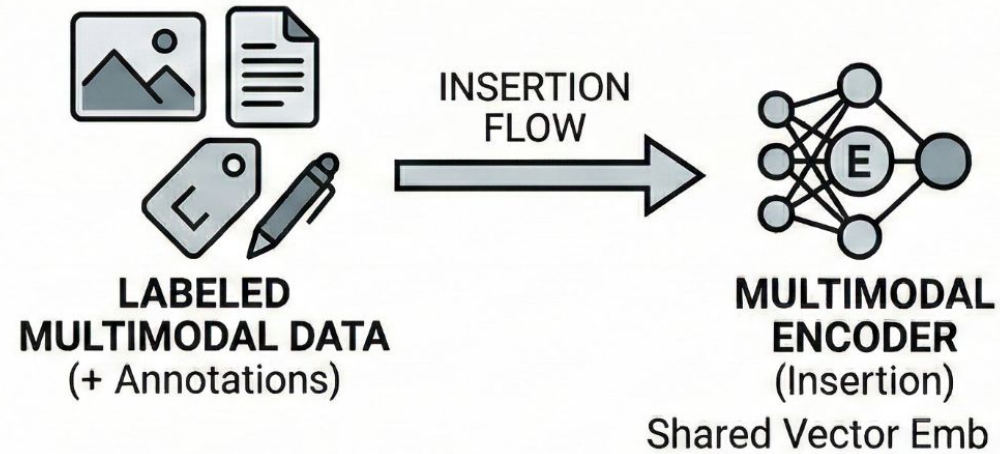
QUERY INFERENCE (SEARCH FLOW)



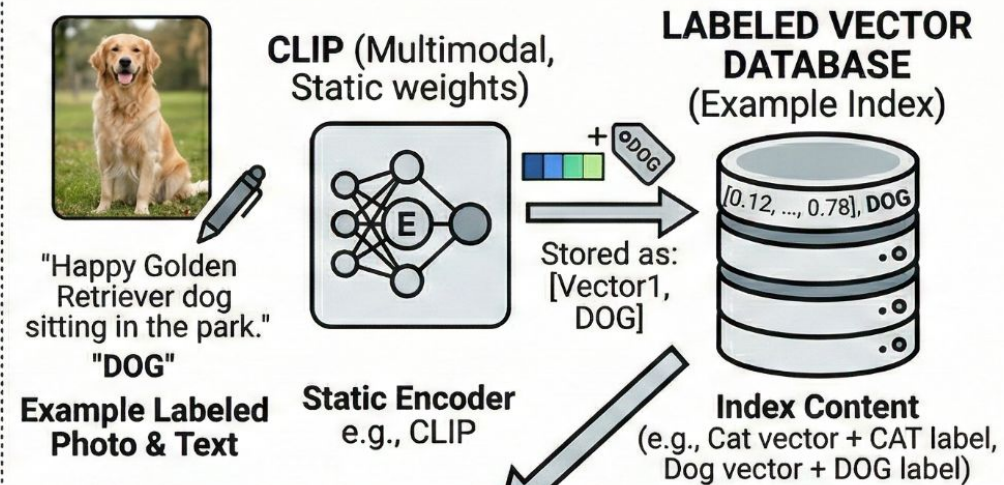
Methodology

A RETRIEVAL-BASED MULTIMODAL LEARNING FRAMEWORK: THEORETICAL METHODOLOGY & CONCRETE EXAMPLE

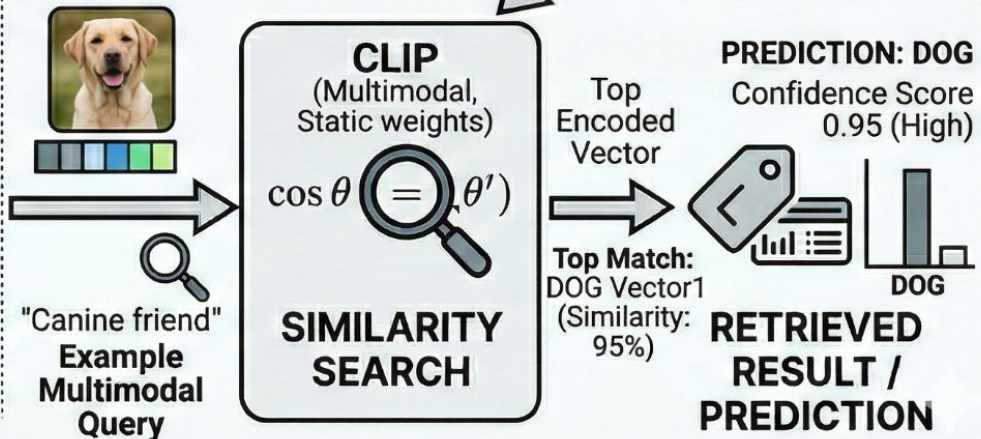
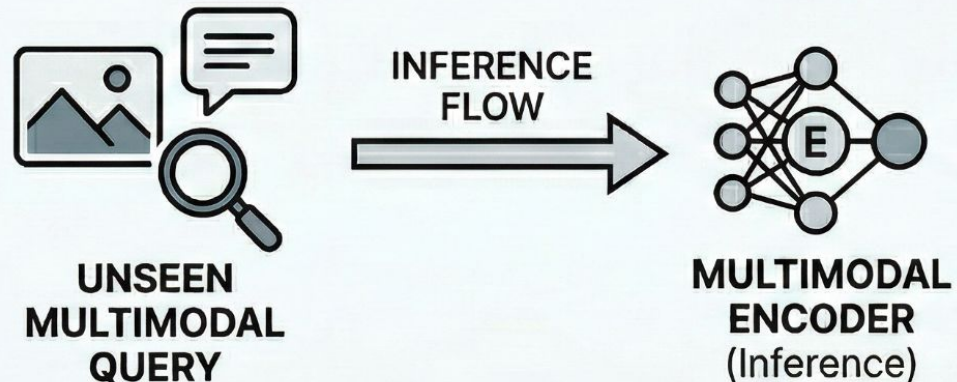
THEORETICAL METHODOLOGY (ABSTRACT)



CONCRETE APPLICATION (EXAMPLE)



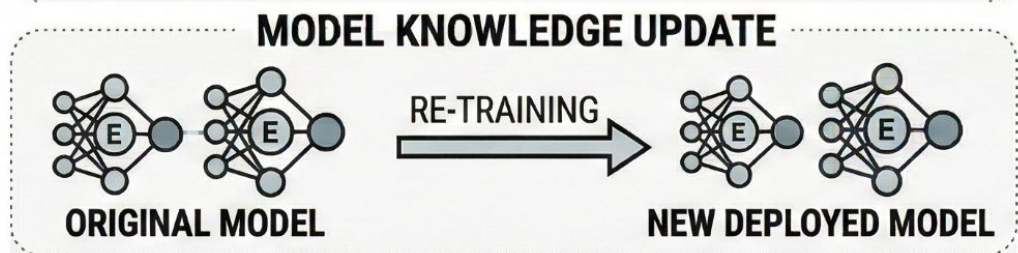
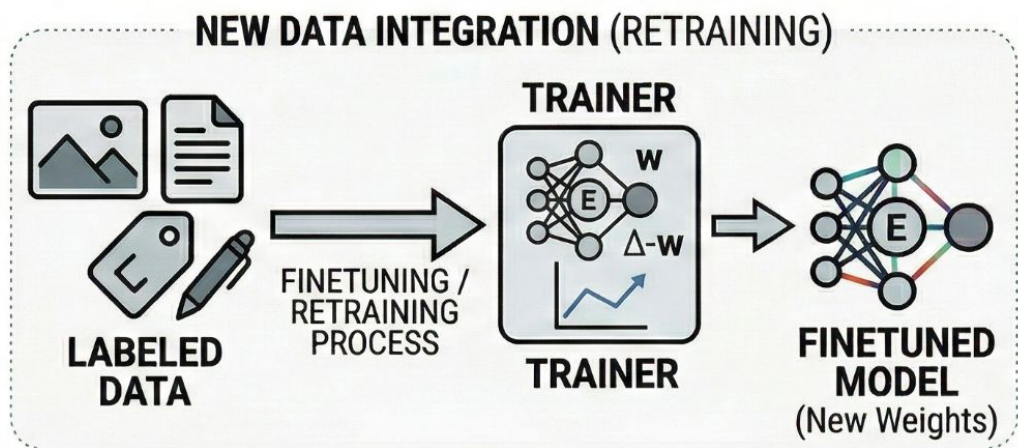
QUERY INFERENCE (SEARCH FLOW)



Methodology

SYSTEM COMPARISON: TRADITIONAL MODEL FINETUNING VS. THE PROPOSED RETRIEVAL-BASED FRAMEWORK (DYNAMIC CONTEXT)

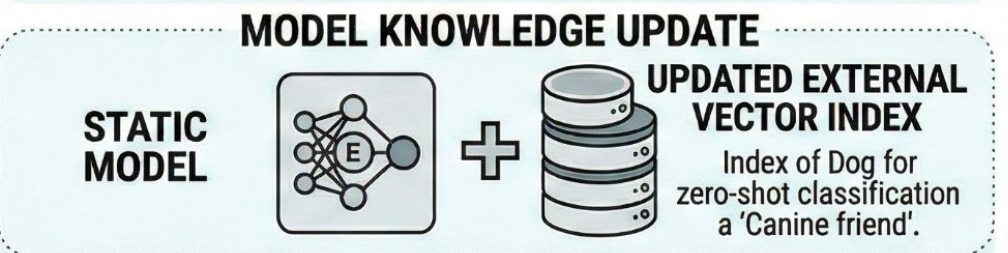
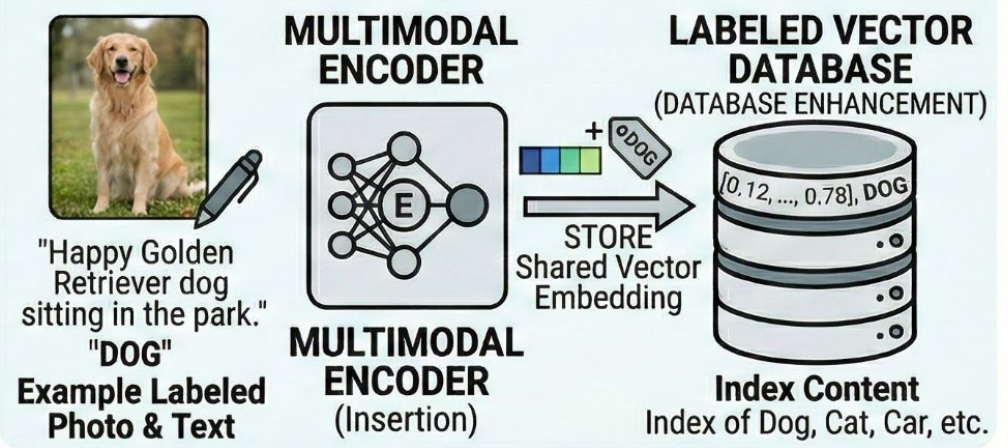
APPROACH A: TRADITIONAL MODEL BUILDING & FINETUNING (Static Model State)



- **KNOWLEDGE STORAGE:** Parametric (Weights)
- **UPDATE SPEED:** Slow (Batch Process)
- **COMPUTATIONAL COST:** High (GPU intensive)
- **FLEXIBILITY:** Low (Difficult to isolate data)
- **SCALABILITY:** Limited by training costs. Updating the entire model state for classification of new products.

↑ ↓	Modified Weights:	New
↓ ⌚	Update Time :	Slow
💰 💰	Computational Cost:	Low
📁 🗃️	Data Management:	High
🌐 📈	Scalability :	Highly Scalable

APPROACH B: THE PROPOSED RETRIEVAL-BASED FRAMEWORK (Dynamic External Context)



- **KNOWLEDGE STORAGE:** Non-Parametric (Exctor Index)
- **UPDATE SPEED:** Real-Time (Instant addition)
- **COMPUTATIONAL COST:** Low (Search intensive, not training intensive)
- **FLEXIBILITY:** High (Easy to add/remove specific data)
- **SCALABILITY:** Highly Scalable

References

- Banani, Y., Agarwal, R., & Almahairi, A. (2024, May). Retrieval-Enhanced Contrastive Vision-Text Models. In *12th International Conference on Learning Representations (ICLR)*. OpenReview.
- Zhang, R., Fang, R., Gao, P., Zhang, W., Li, K., Dai, J., Qiao, Y., & Li, H. (2022, October). Tip-Adapter: Training-free adoption of CLIP for few-shot classification. In *European Conference on Computer Vision (ECCV)* (pp. 493-510). Springer.
- Yasunaga, M., Aghajanyan, A., Shi, W., James, R., Leskovec, J., Liang, P., Lewis, M., Zettlemoyer, L., & Yih, W. (2023, July). Retrieval-Augmented Multimodal Language Modeling. In *International Conference on Machine Learning (ICML)* (pp. 39755-39769). PMLR.
- Gui, Z., Sun, S., Li, R., Yuan, J., An, Z., Roth, K., Prabhu, A., & Torr, P. (2024, June). kNN-CLIP: Retrieval enables training-free segmentation on continually expanding large vocabularies. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE.
- Wang, Z., Liang, J., Sheng, L., He, R., Wang, Z., & Tan, T. (2024, May). A Hard-to-Beat Baseline for Training-free CLIP-based Adaptation. In *12th International Conference on Learning Representations (ICLR)*. OpenReview.
- Li, Y., Guo, J., Qi, L., Li, W., & Shi, Y. (2024, February). Text and Image Are Mutually Beneficial: Enhancing Training-Free Few-Shot Classification with CLIP. In *Proceedings of the AAAI Conference on Artificial Intelligence* (pp. 13695-13703). AAAI.
- Wang, X., Wang, Y., Zhang, Z., & Zhang, Y. (2024, June). Proto-Adapter: Efficient Training-Free CLIP-Adapter for Few-Shot Image Classification. In *Sensors*, 24(11) (p. 3624). MDPI.
- Mitra, K., Santosh, A. V., Teotia, S., & Malapati, A. (2023, December). Effect of Pivot Language and Segment-Based Few-Shot Prompting for Cross-Domain Multi-Intent Identification in Low Resource Languages. In *Proceedings of the 20th International Conference on Natural Language Processing (ICON)* (pp. 235-244). NLP Association of India.
- Baumann, N., Diaz, J. S., Michael, J., Netz, L., Nqiri, H., Reimer, J., & Rumpe, B. (2024, March). Combining Retrieval-Augmented Generation and Few-Shot Learning for Model Synthesis of Uncommon DSLs. In *Modellierung 2024-Workshopband* (pp. 1-15). Gesellschaft für Informatik.
- Ashok, A., & Lipton, Z. (2024, November). Improving Few-Shot Cross-Domain Named Entity Recognition by Instruction Tuning a Word-Embedding based Retrieval Augmented Large Language Model. In *arXiv preprint arXiv:2411.00451*. Cornell University.

List of Publication

1. “An Interpretable ML Framework for Predicting Structural Heart Disease from ECG Using LLM-Powered Insights” (posted on SSRN, Dec 15, 2025):

This research develops and validates a **machine learning framework** to predict **structural heart disease (SHD)** using *electrocardiogram (ECG)* data. The study combines traditional ML models (like logistic regression, support vector machines, and K-nearest neighbors) with a novel **explainability layer powered by large language models (LLMs)**. The LLM component translates complex model results into **clear clinical recommendations**, improving interpretability for use in real-world health care settings. Using a large dataset of 100,000 ECG records, the framework achieved **high predictive accuracy** while enabling actionable insights for clinicians.

Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5875405

List of Certifications

- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI
- **AWS Academy Machine Learning Foundations** – Amazon Web Services (AWS)
- **Data Science Certificate Program** – Ostad
- **Data Science Fundamentals** – Ostad
- **Data Analytics with Python** – Ostad
- **Basic Statistics for Data Science** – Ostad
- **Galactic Problem Solver** – NASA - National Aeronautics and Space Administration
- **Workshop on Google Data Studio** – Data Science Lab-DIU

THANK YOU